

PROCESS SPECIFICATION

PS-2024-011 Rev C

Heat Treatment of Steel and Stainless Steel Alloys

1. SCOPE

This specification defines heat treatment requirements for carbon, alloy, and stainless steel parts. All heat treatment shall be performed by a NADCAP-accredited facility. Pyrometry shall conform to AMS 2750.

2. APPLICABLE STANDARDS

AMS 2759 — Heat Treatment of Steel Parts, General Requirements

AMS 2750 — Pyrometry

AMS 5643 — Heat Treatment of 17-4 PH Stainless

3. PROCESS REQUIREMENTS BY ALLOY

Alloy	Condition	Temp (°F)	Time (hr)	Quench	Hardness
4340	Normalize	1600 ±25	1 per inch	Air	—
4340	Harden	1525 ±25	1 per inch	Oil	50-55 HRC
4340	Temper	800-1000	2 min	Air	38-42 HRC
17-4 PH	H900	900 ±15	1	Air	40-44 HRC
17-4 PH	H1025	1025 ±15	4	Air	35-39 HRC
316 SS	Anneal	1900 ±25	0.5 min	Water	—

4. RECORDS

Heat treatment records shall include: furnace ID, load thermocouple data, actual time at temperature, quench media temperature, and hardness results. Records shall be retained for a minimum of 10 years.

Effective Date: March 1, 2024 | Approved by: J. Harrison, Quality Eng.